TOMÁŠ HUTŇAN

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Brno, Czechia · Slovak national (EU citizen)

EDUCATION

MASARYK UNIVERSITY | *BSc Informatics* Sep 2024 – Present

- Academic standing: 95th percentile of cohort.
- Relevant coursework: Machine Learning, Statistics, Image Processing, Modelling & Simulation.

EXPERIENCE

MASARYK UNIVERSITY | *Faculty of Informatics* Brno, Czechia

Research Assistant Oct 2026 – Present

- Research LLM safety and clinical guardrails within Project EMPOWER.
- Generate synthetic oncology datasets to find where models fail.

Teaching Assistant Sep 2025 – Present

- Teach weekly seminars in Intro to AI, Algorithms, and Programming for 20+ students.
- Review 100+ submissions weekly, coaching students on maintainable software practices.

CHYRONHEGO CZECH S.R.O. | *Broadcast Graphics*

Software QA & Automation Intern Nov 2024 – Dec 2025

- Built Python test automation for high-availability systems (SQL, APIs, Docker).
- Authored 1,000+ automated test cases, replacing manual regression testing.
- Reviewed 100+ pull requests and proposed code improvements adopted into production.

SELECTED PROJECTS

METHANE SEGMENTATION ON FPGA | *Researcher · Edge ML* Jun 2026 – Present

- Research on-board methane segmentation for satellites to cut downlink data.
- Compare PTQ vs. QAT INT8 models on a ZCU104 FPGA (Vitis AI) against low-power CPUs.

CUDA IMAGE ENHANCEMENT | *GPU Engineer · C++ / CUDA* Mar – May 2026

- Ported TESCAN's CPU microscopy pipeline to CUDA: ~10 s/image to 25+ FPS on a laptop GPU.
- Optimized denoising and deconvolution kernels; validated output with PSNR/SSIM.

TEARZ | *Project Lead · ML / Computer Vision* Apr 2026

Honorable Mention, UPJŠ Challenge – Hack Košice 2026

- Led a team screening tear microscopy images for eye and systemic diseases.
- Built a CV feature pipeline (93% F1) and a ResNet50 ensemble with leak-free splits.

STOCKMATICS | *ML Engineer · Python / NLP* Mar 2025

1st Place Overall, Newsmatics Hackathon 2025

- Built a news-driven market analytics platform with a sentiment-based Fear & Greed index.
- Measured consensus across news outlets using transformer embeddings.

AWARDS

Hackathons Tatra Banka Challenge winner (2024, 2026), 3rd Place Hack Košice 2023, 3× MLH Best Use
Game Dev 2nd Place at national competitions: Špongia 2021, iHRA 2022, iHRA 2023

SKILLS

Core Stack Python, C++, PyTorch, CUDA, Docker, Linux, OpenCV, Pandas
Also Used TypeScript, C#, FastAPI, PostgreSQL, MongoDB, React
Specializations Computer Vision, NLP, GPU Optimization, Automated Testing
Languages Slovak (Native), English (C1/C2), German (A1/A2)